

Unit 4, Lesson 10: Using Circumference and Area to Solve Problems

Benchmarks

MA.7.GR.1.3 Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

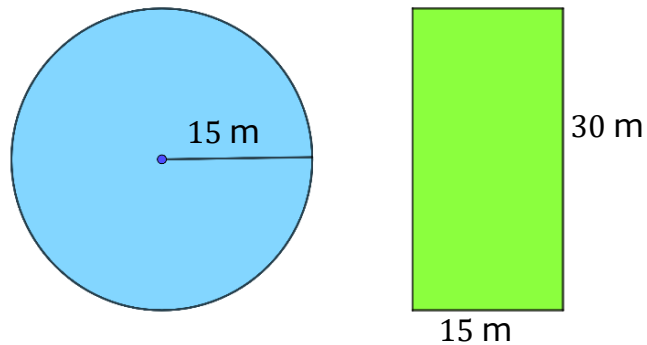
MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.

Learning Target

- ☐ I can use the circumference and area of a circle to solve problems.

4.10.1 Warm-Up: Would You Rather?

A circle with a radius of 15 meters (m), and a rectangle with a length of 30 m and width of 15 m are shown.



1. Explain if you would rather run the circumference of the circle or the perimeter of the rectangle.

4.10.2 Activity: Sage and Scribe

For the first problem, decide who will be the Sage and who will be the Scribe.

While working a problem, the Sage is the only one who can talk. The Scribe must write down only what the Sage says to solve the problem.

For each problem, swap who is the Sage and who is the Scribe.

Problem A	
Name of Sage: _____	A lawn sprinkler sprays water in a circle with a radius of 6.27 feet (ft).
Name of Scribe: _____	
	How much of the lawn will be watered from the sprinkler? Use $\frac{22}{7}$ for π .

Problem B

Name of Sage:

Name of Scribe:

Felicia has a circular rug that measures $5\frac{7}{12}$ feet (ft) from the center to the edge of the rug.



Felicia needs to replace the fringe on the edge of the rug after her dog chewed it. How much fringe does Felicia need to buy? Use 3.14 for π .

Problem C

Name of Sage:

Name of Scribe:

The Wheel at ICON Park is approximately 400 feet (ft) tall.



If a person takes one trip around the Ferris wheel, about how far will the person have traveled? Use $\frac{22}{7}$ for π .

Problem D

Name of Sage:

Name of Scribe:

William is painting a large image of the full moon in the image below on his wall. He wants the painting of the full moon to cover his wall from the floor to the ceiling.



If the wall is 3.048 meters (m) tall, how much of the wall will be covered in paint? Use 3.14 for π .

4.10.3 Your Turn!

1. Yvonne's company wants to put their logo on frisbees to give away at a convention.

If each frisbee has a diameter of $12\frac{1}{2}$ inches (in.), and the logo can cover the entire top of a frisbee, how much space is available for the logo on each frisbee? Use $\frac{22}{7}$ for π .



2. A jeweler is asked to wrap a rare coin with gold braid to make a pendant.

If the coin has a diameter of 0.75 inches (in.), what is the length of the gold braid the jeweler will need? Use $\frac{22}{7}$ for π .



4.10.4 Wrap-Up: Reflection

Today I deserve



for ...

Today I



by doing ...

Today I was



by ...